



DISTRIBUTION AND DENSITY OF LOBSTERS IN THE BRAZILIAN OCEANIC ECOSYSTEM ROCAS ATOLL

BY

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ABSTRACT

Lobsters are recognizable faunal elements that play an important role as top predator in the trophic webs in benthic ecosystems and have an economic importance due to the intensive and valuable fishery. In Rocas Atoll (03°51'S 33°48'W) the presence of five species of lobsters in low tide pools was observed by visual census. These were: *Enoplometopus antillensis* Lütken, 1865; *Palinurellus gundlachi* Von Martens, 1878; *Panulirus argus* (Latreille, 1804); *Panulirus echinatus* Smith, 1869; and *Parribacis antarcticus* (Lund, 1793). This atoll appears to be dominated by *P. echinatus* followed by *P. argus*, *P. antarcticus*, *P. gundlachi* and *E. antillensis*, respectively. We also observed the presence of some potential predators that perhaps feed on lobster species and control these populations in the atoll. This work and future information could help to better understand the variability of lobster diversity and density in this unique atoll.

Key words. — Lobster diversity, Rocas Atoll, visual censuses

RESUMO

As lagostas são habitantes reconhecíveis que desempenham um importante papel como principais predadores nas redes tróficas em ecossistemas bentônicos e têm uma importância econômica devido à intensa e valiosa pesca. No Atol das Rocas (03°51'S 33°48'W) foi observada por meio de censos visuais a presença de cinco espécies de lagostas nas piscinas formadas na maré baixa. Foram elas: *Enoplometopus antillensis* Lütken, 1865; *Palinurellus gundlachi* Von Martens, 1878; *Panulirus argus* (Latreille, 1804); *Panulirus echinatus* Smith, 1869 e *Parribacis antarcticus* (Lund, 1793). Este atol é dominado por *P. echinatus* seguido por *P. argus*, *P. antarcticus*, *P. gundlachi* e *E. antillensis*, respectivamente. Também observamos a presença de alguns potenciais predadores que talvez se alimentem de espécies de lagostas e controlem essas populações no atol. Este trabalho e informações futuras podem ajudar a entender melhor a diversidade de lagostas e a variabilidade da densidade destes organismos neste atol único.

Palavras-chave. — Diversidade de lagostas, Atol das Rocas, censos visuais

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INTRODUCTION

Lobsters play an important role as top predators in the trophic webs in benthic marine ecosystems and also have an economic importance due to the intensive and valuable fishery (Cruz et al., 2015). Despite the ecological and economic importance of the spiny lobster for many countries along the Atlantic Ocean, just a few detailed studies for specific species have been conducted to clarify variations observed in recruitment, mechanisms responsible for larval transport, and their relation to adult stocks (Cruz et al., 2013, 2014, 2015).

The most studied lobster in Brazil is *Panulirus argus* (Latreille, 1804), which is the most abundant and consequently also the most exploited by the fishery. Studies exist that address various aspects of its biology, like density, sex ratio, habitat, relation predator/prey, and other features of this species in the North Atlantic Ocean. With regard to Brazil, some studies on fishery and aquaculture have been performed (Coelho et al., 1996; Fonteles-Filho, 1997; Carvalho & Ogawa, 2001; Fonteles-Filho, 2005, 2007) and also a number of larval ecological contributions have recently been published (Silva et al., 2001, 2013; Cruz et al., 2013, 2014, 2015).

Nonetheless, ecological studies about other lobster species are scarce or absent, even though these species are caught incidentally during the regular spiny lobster fisheries that are targeted primarily at *P. argus*. Studies on subjects such as habitat structure of lobsters and their density in situ are scarce and rare for the Brazilian shelf and the country's oceanic islands (Oliveira et al., 2008; Faria-Júnior et al., 2013; Gaeta et al., 2015).

In Rocas Atoll, there are records of six lobster species: *Panulirus echinatus* Smith, 1869, *P. argus*, *Panulirus laevicauda* (Latreille, 1817), *Parribacus antarcticus* (Lund, 1793), *Enoplometopus antillensis* Lütken, 1865, and *Palinurellus gundlachi* Von Martens, 1878 (cf. Gaeta et al., 2015). Some aspects about adult population structure in *P. echinatus* were discovered by Silva et al. (2001) with regard to this atoll.

The distributions of those lobsters species are variable in the Atlantic Ocean; *Enoplometopus antillensis* is a species of reef lobster endemic to the Atlantic Ocean that is found on rocky and coral reefs, where it hides in small crevices (Chan & Wahle, 2011). This species is found in Bermuda, Florida, the Bahamas, Bonaire, Panama, Venezuela, Brazil (off the northeast coast and off Rio Grande do Norte State and Rocas Atoll as well as in the Fernando de Noronha Archipelago) (Alves et al., 2008; Gaeta et al., 2015), on Ascension and the St. Helena Islands, Madeira, the Canary Islands, the Gulf of Guinea, and the Cabo Verde Islands (Poupin, 2003).

Palinurellus gundlachi is rather scarce and it is found in a depth range from 1.5-35 m in rather inaccessible places among rock and coral, and it is distributed

from Bermuda, the Bahamas, southern Florida, Yucatán, the Caribbean Arc, Curaçao and Brazil (the states Pernambuco and Bahia, and Rocas Atoll) (Carpenter, 2002; Gaeta et al., 2015).

Panulirus argus is found among rocks, on reefs, in eelgrass beds or in any habitat that provides protection. It inhabits shallow waters, occasionally down to 100 m depth or even deeper, as in the Brazilian waters (Holthuis, 1991; Cruz et al., 2011). This species occurs from Florida to southern Brazil. Along the Brazilian coast, it occurs from Pará (01°27'S 48°30'W) to São Paulo (23°32'S 46°38'W) and on two assemblages of oceanic islands: the Fernando de Noronha Archipelago (03°50'S 32°24'W) and Rocas Atoll (03°45'S 32°19'W) (Holthuis, 1991; Coelho & Ramos-Porto, 1998; Melo, 1999).

Panulirus echinatus inhabits a depth range of 0-35 m, but usually not deeper than 25 m (Holthuis, 1991). This species had been recorded mainly on islands in the Central Atlantic (the Canary Islands, the Cabo Verde Islands, Ascension, and St. Helena) and along the northeastern Brazilian coast (Ceará to Rio de Janeiro, including São Pedro and São Paulo [= St. Pauls Rocks], and the Fernando de Noronha Archipelago, Rocas Atoll, and Trindade Island) (Manning & Chace, 1990; Coelho & Ramos-Porto, 1998; Melo, 1999).

Panulirus laevis is usually found in coastal waters, down to 50 m depth on rock or coral substrate (Holthuis, 1991) and occurs from Bermuda and Florida to Brazil (Fernando de Noronha Archipelago, Rocas Atoll and from Maranhão to Rio de Janeiro (03°50'S 38°20'W to 23°17'S 44°10'W)) (Holthuis, 1991; Coelho & Ramos-Porto, 1998; Melo, 1999).

Parribacus antarcticus is found at depths from 0-20 m on coral or stone reefs with a sandy bottom (Holthuis, 1991) and perhaps also found at other depths (Silva et al., 2013). This is a world-wide occurring species and it is known from seamounts from Florida to Brazil, and along the Brazilian coast from Amapá to São Paulo (00°02'N 51°03'W to 23°32'S 46°38'W), including Rocas Atoll and the Fernando de Noronha Archipelago (Holthuis, 1991; Coelho & Ramos-Porto, 1998; Melo, 1999).

As a consequence, in the present study we gathered information on distribution and density of all lobsters across the various habitats in the ecosystem of Rocas Atoll, meant to constitute an initial step toward identification of the population structure and habitat use of the several lobster species there present.

MATERIAL AND METHODS

Site description and habitat

Rocas Atoll (03°51'S 033°48'W) (fig. 1) is the only atoll in the South Atlantic and one of the smallest in the world (Silva et al., 2001). In 1979, this atoll was the

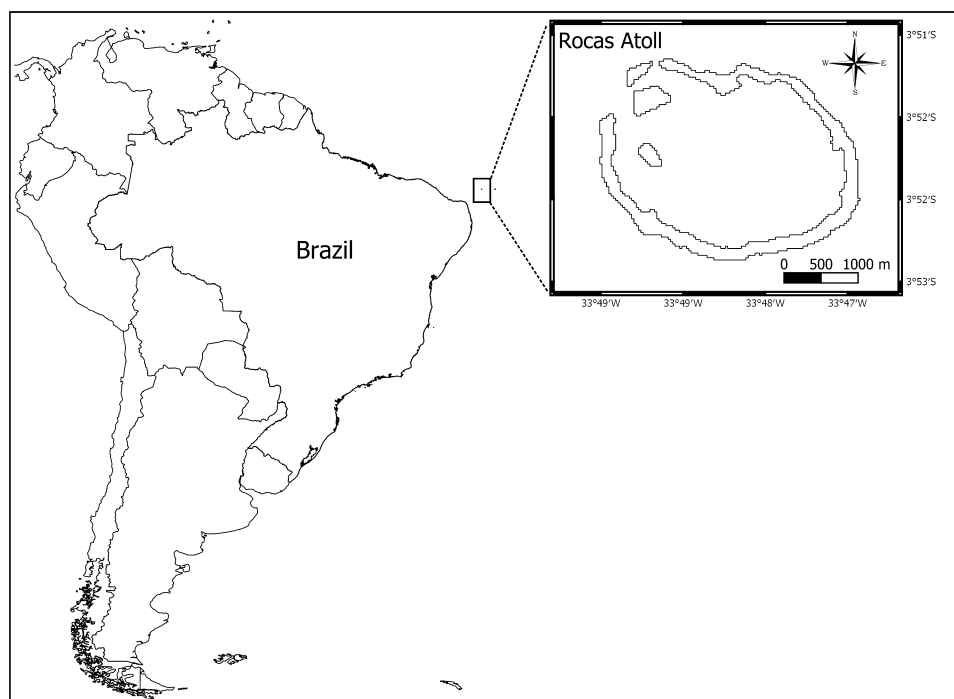


Fig. 1. Location of the Rocas Atoll on the Brazilian coast.

first area to be established as a Marine Protected Area in Brazil. This and other oceanic islands along the coast of Brazil have great importance for the territorial policy of the Federation, due to their remote location off the continental coast, resulting in an increase of the Brazilian Exclusive Economic Zone.

The Marine Reserve includes an area of 360 km², located 266 km off the nearest mainland coast (Pereira et al., 2013). The tidal regime in Rocas is semi-diurnal and mesotidal, with a maximum height of 3.8 m (Gherardi & Bosence, 2001) and the sea surface salinity varies from 36 to 37 (Gherardi & Bosence, 1999). The mean temperature of the sea water varies from 26°C to 28.3°C (Hoflich, 1984; Servain et al., 1987) but it could increase until 42°C in the atoll's inner rock-pools (Soares et al., 2009). Wave action is concentrated in the southeastern portion of the atoll, although the refraction of waves caused by the sea mountain that supports Rocas produces braking waves that also act on the western and southwestern portions (Gherardi & Bosence, 2001).

Rocas atoll is composed of *Homotrema rubra* (Lamarck, 1816), which is an encrusting foraminiferan, acervulinids (also encrusting Foraminifera), corals, tubes of polychaete worms, molluscs, vermetid gastropods, sponges, coralline algae, cemented sediments, and growth cavities (Gherardi & Bosence, 2001). Some of these organisms, like foraminiferans, corals, molluscs, and coralline algae, are

cited to contribute to the generation of biogenic particles in shallow tropical reef systems (Weber & Woodhead, 1972).

Sediments deposited in the interior of atoll reefs consist almost entirely of skeletal carbonate and the development of a sedimentary facies is mediated by the interplay between the sediment source supply, its physical properties, and several processes that redistribute and settle the sediments (Maxwell et al., 1964; Milliman, 1974). The predominant sediment inside the atoll is coarse sand followed by medium sand and inside the tide pools it ranged from coarse to very coarse sediment, which can be explained by the high hydrodynamism that results in insufficient mud production and consequently the absence or only a low amount of fine particles, such as mud, in this atoll (Pereira et al., 2013).

Data collection

The data were obtained in the pools formed during low tide and on the sandy island beaches at Rocas Atoll (3°49'S 33°4'W) during expeditions in April and October 2013 and April and October 2014. Lobsters were recorded at depths of 0 to 6 m during non-destructive visual censuses with minimal ecosystem interference.

Data collection was done by two divers snorkelling inside each tide pool, searching for lobsters in crevices, shelters, holes and under slabs during daytime. The presence of some predators is recorded but unfortunately it was not possible to analyse the stomach contents of those predators, because it would then have been necessary to sacrifice some specimens, and it is not recommended to do so in a Marine Protected Area as this atoll. Some data were collected around the two islands when lobster specimens were found dead in the sand. In addition, one night dive and one night walk were performed in order to make also some nocturnal observations, but it must be stated that just this one dive was possible, because of logistic and tidal conditions.

All specimens were identified in situ and, if necessary, photographed for further identification. All identifications were based on the lobster key of Holthuis (1991) and the main characteristics, i.e., colour, spine patterns, antenna size, and other features were used to verify the morphology of each species. The carapace lengths of lobsters were not evaluated for this study but specimens' sizes were determined as Carapace Length for males and females, starting at 38 mm up to and including 119 mm.

A lobster species distribution map for Rocas Atoll was thus made, indicating the presence or absence of each species in each tide pool inside the atoll. There is, unfortunately, no perfect method to determine the abundance of animals in the sea (Lessios, 1996). However, as an approximation, the abundance index can be given by the equation $N = n/A$ (cf. Seber, 1982), where n is the number

of individuals and A the area of the habitat; yet, this approach imposes some restrictions on the investigators, viz., through adversities in the environment as well as by a possible migratory pattern of the target species. Therefore, it is more appropriate to determine the population density per area, just like indices of relative abundance. So, the response variable produced from the area was the density (lobster number/m²) per species.

Analyses were made to evaluate data normality through the Shapiro-Wilk test. Density was evaluated per species between months and between tide pools by a two-way ANOVA and in case of differences an a posteriori Tukey test was done to verify which those differences were (Zar, 2010).

Spatial distribution and area explored in Rocas Atoll

The mapping of the target area was based on two images of the Geomorphological map (modified from Pereira et al., 2013) and one image of the lobster distribution on benthic habitats. Using the methodology of Cerdeira-Estrada et al. (2008), each image was geometrically corrected to optimize precision, selecting and locating four control positions evenly distributed on the image. Coordinates for each point were obtained from topographic maps on a 1 : 40 000 scale. Geodetic Datum GWS84 (Rocas Atoll, Brazil) was obtained from the Clarke 1866 spheroid (an ellipsoid model to determine the figure of the Earth) and the cartographic projection was done using Lambert conformal conic projection, that is a conic map projection with straight-line meridians that meet at a common center beyond the limits of the map and with parallels of which two are standard that are arcs of circles intersecting the meridians at right angles, which is used for aeronautical charts and mapping systems (Cerdeira-Estrada et al., 2008).

RESULTS

Lobster distribution and behaviour

In total, 15 tide pools (P1 to P15) were sampled, as well as one very shallow, rocky slab (R) near Cemitério Island (fig. 2). The mean depth and the respective confidence interval (CI) were 2.52 ± 0.4 m ($p < 0.05$) for the sampled periods, but depth actually ranged from 0.3 to 6.0 m. Depth may vary, however, according to the sediment movement inside the atoll along and between years. The mean temperature and the respective confidence interval were $28.11 \pm 0.65^\circ\text{C}$ ($p < 0.05$) for the sampled periods, and temperature ranged from 27 to 31°C.

We found a total of five species of lobsters in the atoll: *Panulirus echinatus*, *Panulirus argus*, *Parribacus antarcticus*, *Palinurellus gundlachi*, and *Enoplometopus antillensis*.

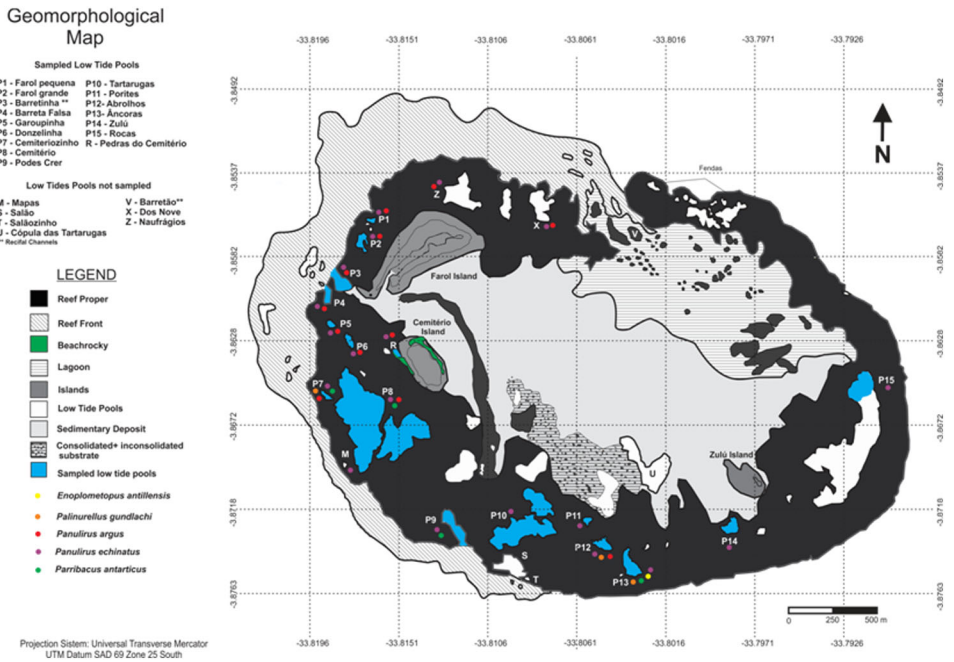


Fig. 2. Geomorphological map of Rocas Atoll with the distribution of the lobster species in the sampled tide pools on the atoll. [Modified from Pereira et al., 2013.]

On all expeditions, *P. echinatus* was the more representative species (96.46%) followed by *P. argus* (3.37%), *P. antarcticus* (0.1%), *P. gundlachi* (0.06%) and *E. antillensis* (0.01%). The more abundant and dominant species in the atoll thus is *P. echinatus* and it is well distributed in Rocas Atoll, occurring in all tide pools and on the beach rocky slab (see purple dots in fig. 2).

The spiny lobster *P. argus* was observed predominantly on the west side of the atoll, highlighting the northwestern pools (see red dots in fig. 2) that had the greatest density of this species ($0.00465 \text{ ind.m}^{-2}$) in P2. *P. antarcticus* appeared with a total of seven individuals located in only three tide pools (P7, P9 and P13) during visual censuses (see green dots in fig. 2). During the day it is very difficult to see this species, because it is very similar in colour and structure to the walls and ceilings of caves and shelters. At night, we could find more than 5 specimens in less than 10 minutes, just by walking in the sand and without actively searching it.

The other two species recorded are known to be rare, and thus only one individual of *E. antillensis* was recorded in pool P13 (see yellow dot in fig. 2), as well as four specimens of *P. gundlachi*, scattered in tide pools P7, P12 and P13 (see orange dots in fig. 2). Two of the pools, P7 and P13, were the most diverse among the fifteen pools sampled, with the presence of four lobster species each.

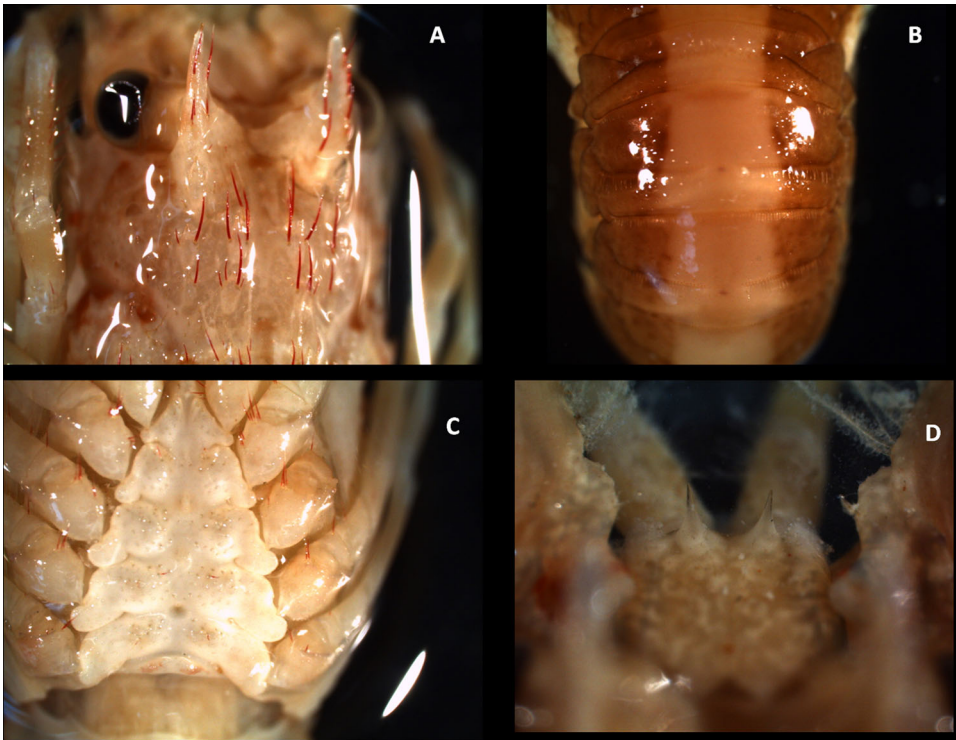


Fig. 3. Morphological characteristics of *Panulirus echinatus* Smith, 1869 post-puerulus. A, Red spines distributed throughout the body, mainly on the carapace; B, abdominal grooves; C, punctual depression on the thoracic sternum; D, presence of two spines on the antennular plate.

We observed all life stages (post-puerulus (recruits), juveniles, and adults) of *P. echinatus*, some of them were copulating and some breeding females were visualized in their shelters. These observations thus corroborate that there is an established and healthy population of *P. echinatus* in Rocas Atoll. The postlarva has red spines, distributed throughout the body, but mainly on the carapace. Abdominal grooves are present as well as a punctual sinking in the thoracic sternum. Another distinguishing characteristic is the presence of two spines on the antennular plate (fig. 3).

Gregarious behaviour was observed in *P. echinatus* shelters, i.e., in crevices and on slabs. Most of those groups counted more than 10 individuals, and some specific shelters were occupied by 40 individuals or more. Moreover, these specimens were mostly of small and medium size, with just a few larger animals cohabiting in the shelter. Most of the time, the first lobster in these specific shelters to make contact or interact with observers, was a large individual, most frequently a male. In general, the grouping or flocking of individuals constitutes an anti-predatory behaviour and it is used by several species.

Density of spiny lobsters

The more representative densities were recorded for *P. echinatus* and *P. argus* followed by *P. antarcticus*, *P. gundlachi* and *E. antillensis*, respectively. The mean $\pm$ CI (95%) density for *P. echinatus* in Rocas Atoll was 0.032 ± 0.007 ind.m⁻², while for *P. argus* it was 0.00093 ± 0.00056 ind.m⁻², *P. antarcticus* was represented by 0.000021 ± 0.000025 ind.m⁻², *P. gundlachi* counted 0.000019 ± 0.000024 ind.m⁻², and *E. antillensis* was present with 0.000002 ± 0.000035 ind.m⁻². The density of *P. echinatus* is approximately 34 times as high as that of *P. argus*.

The highest densities of both species were found in the northwestern pools of the atoll. *P. argus* were recorded in low densities coexisting with the congeneric *P. echinatus* in the same shelters, caves, and holes in those northwestern pools, yet, there was no relation between the densities of *P. argus* and *P. echinatus*. This coexistence, thus, may be indicating a shortage of refuges in this particular environment.

The tide pools P4, P5, P6 and the slab R were the more abundant sites for *P. echinatus* (fig. 4) whereas P2, P3 and again R were the more abundant pools for *P. argus* (fig. 5). The ANOVA shows no statistical differences between months for *P. echinatus* and *P. argus*, so we opt to use the density mean for each species. However, differences between tide pools were indicated by the ANOVA and an a

Geomorphological Map

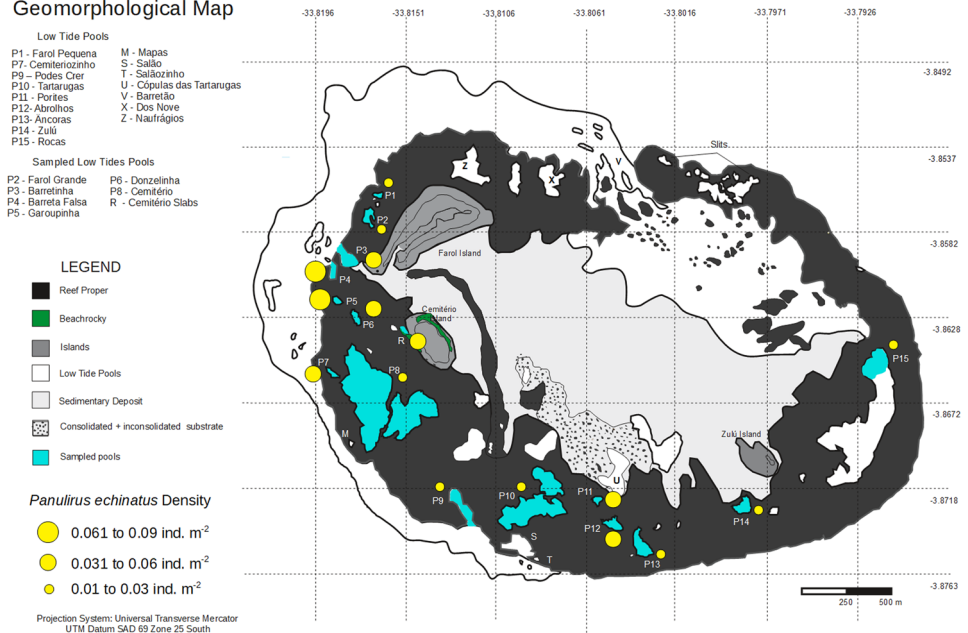


Fig. 4. *Panulirus echinatus* Smith, 1869 density (ind.m⁻²) on Rocas Atoll.

Geomorphological Map

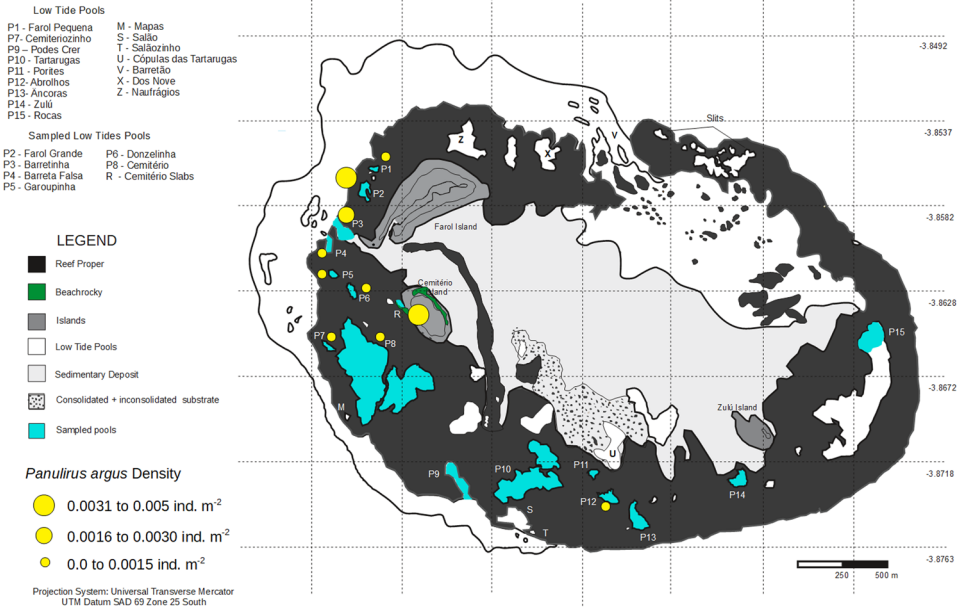


Fig. 5. *Panulirus argus* (Latreille, 1804) density (ind.m⁻²) on Rocas Atoll.

posteriori Tukey test was done to specify those differences for *P. echinatus* and for *P. argus*, that have subsequently been represented in a density map for each species (figs. 4 and 5). For *P. antarcticus*, *P. gundlachi* and *E. antillensis* no data analysis was possible, because of very low densities or absence of these species in many sampled areas.

During the study, specifically in the manual collection of the lobsters, we had to pay close attention to capture and return the animals in time while also measuring them, because there was always a *Lutjanus jocu* (Bloch & Schneider, 1801) lurking around in search of an easy prey, like lobsters out of their refuges. Evidence from the diving observations has shown the presence of potential predators in the sampled tide pools, and the most frequent predators were *Lutjanus jocu* and *Ginglymostoma cirratum* (Bonnaterre, 1788). The rare presence of octopuses (*Octopus insularis* Leite & Haimovici in Leite, Haimovici, Molina & Warnke, 2008, *Octopus hummelincki* Adam, 1936 and *Octopus defilippi* Vérany, 1851 [currently known as *Macrotritopus defilippi* (Vérany, 1851)]), sharks (*Negaprion brevirostris* (Poey, 1868)) and rays (*Dasyatis americana* Hildebrand & Schroeder, 1928), and various species of bony fishes (*Epinephelus itajara* (Lichtenstein, 1822) and *Mycteroperca bonaci* (Poey, 1860), also including the morays *Gymnothorax miliaris* (Kaup, 1856) and *Gymnothorax funebris* Ranzani, 1839) were also found in the pools, but not as frequently as compared with *Lutjanus jocu* and *Ginglymostoma cirratum*.

DISCUSSION

Spiny lobsters are nocturnal animals that strongly depend on daytime shelter for survival. The tide pools are shallow waters that support a great biodiversity in the reef construction that promotes shelter and refuges for many different species. Habitat complexity has also been shown to influence the distribution and abundance of a diverse group of mobile animals (Hacker & Steneck, 1990; O'Conner, 1991; Schneider & Mann, 1991).

Experimental habitat manipulations demonstrate a positive relationship between prey survival and structural complexity of the habitat (Heck & Crowder, 1991). For species that demonstrate both shelter-seeking and gregarious behaviour as lobsters, shelter preferences and the resultant survival rates may differ not only with shelter features, but also with the individual's body size and group size or behaviour. The gregarious behaviour normally is a result of a lack of shelters for every animal, and also an anti-predatory behaviour.

The predominance of *Panulirus echinatus* was expected, because this species is preferentially distributed on shallow reefs in oceanic islands (Holthuis, 1991). In addition, we hypothesize that there also is a relation with larval recruitment and settlement, since the high abundance of *P. echinatus* in this area probably indicates that certain environmental factors favour the establishment of this species. This study has not specifically studied the movements of recruitment and settlement, but we observed that densities inside the tide pools do not differ statistically between months or years, which may indicate a residential behaviour. The great abundance of this population may provide spawned larvae for self-recruitment and also be a source for *P. echinatus* populations elsewhere. We also hypothesize that the lobsters are ecologically trapped in Rocas Atoll, because they cannot migrate from there by walking, due to the huge depth around this atoll and the oceanic circulation pattern around the island. As a result, the only way to get into and out of this atoll is by larval recruitment. Therefore, a study is being conducted in this area to evaluate the recruitment patterns and genetic flux between this and other *P. echinatus* populations in the Atlantic Ocean.

The species *Panulirus argus* was recorded in low densities, coexisting with the congeneric *P. echinatus* in the same shelters, caves and holes in the northwestern pools inside the atoll. Den sharing by conspecifics is common in lobsters (Cobb, 1981; MacDiarmid, 1994; Butler & Herrnkind, 2000). In Mexico, congeneric cohabitation of lobsters was found to be independent of lobster and predator densities, and more individuals of *P. argus* were found sharing dens with *Panulirus guttatus* (Latreille, 1804) (cf. Briones-Fourzán & Lozano-Álvarez, 2008). Those authors related that it may simply reflect the stronger tendency of *P. argus* juveniles to aggregate, irrespective of the presence of other species. We found the same

pattern in Rocas Atoll, i.e., *P. argus* aggregating in dens regardless of the presence of *P. echinatus*, rather than *P. echinatus* aggregating regardless of *P. argus*. The fact that lobsters are trapped in this atoll makes the coexistence an obligation, because of the great abundance of *P. echinatus* and *P. argus* in a limited space inside the tide pools. Probably, the local coexistence of these related species is mediated by a differential use of resources at a particular scale indicating niche-partitioning, or by ecological trade-offs as suggested by Kneitel & Chase (2004) and Lozano-Álvarez et al. (2007).

In this study, we found a mean density of spiny lobsters higher than in other protected areas such as in Florida and in Belize (table I): it was truly notable that there was a concentrated, high density of lobsters in this atoll, in comparison to other sites. Also in the Caribbean area differences in lobster density were observed between protected and unprotected marine areas, with higher densities in the protected areas (Eggleston & Dahlgren, 2001; Acosta & Robertson, 2003; Eggleston et al., 2008). Probably, if this protected atoll is compared to other unprotected localities along the Brazilian coast, a difference in abundance will be found between them. Another protected area that was sampled in Brazil was Abrolhos Marine National Park (Faria-Júnior et al., 2013) and species densities were lower there than these data from the Rocas Atoll, at least those for *P. argus* and *P. echinatus*. Faria-Júnior et al. (2013) found *P. argus* and *Panulirus laeviscauda* predominating in this locality near the continent and reported the first record for *P. echinatus* in this area, whereas in Rocas Atoll, an oceanic region, we observed the prevalence of *P. echinatus* in coexistence of a few specimens of *P. argus* and the absence of *P. laeviscauda*. The distribution of different lobster species is

TABLE I
Comparison of lobster densities at different sites

Species	Mean density (ind/m ²)	Site	Reference
<i>Panulirus argus</i> (Latreille, 1804)	0.009	Belize Atoll	Acosta & Robertson (2003)
<i>Panulirus argus</i>	0.025	Florida	Eggleston & Dahlgren (2008)
<i>Panulirus argus</i>	0.200	Florida	Eggleston et al. (2008)
<i>Panulirus guttatus</i> (Latreille, 1804)	0.006	Belize Atoll	Acosta & Robertson (2003)
<i>Panulirus guttatus</i>	0.020	Anguilha	Wynne & Côté (2007)
<i>Jasus edwardsii</i> (Hutton, 1875)	0.060	New Zealand	Kelly (2000)
<i>Jasus edwardsii</i>	0.020	New Zealand	Davidson (2002)
<i>Panulirus echinatus</i> Smith, 1869	0.032	Brazil (Rocas Atoll)	This study

consistent with that reported in previous studies on spatial distribution, according to the presence of the preferred habitat for each species (Holthuis, 1991).

In Glover's Reef Atoll (Belize), lobster densities inside and outside a reserve were compared, with as a result that a high abundance was found inside the protected area, as well as some differences between particular types of reef. The mean densities inside that atoll were 4 times lower than those observed in our study of Rocas Atoll (table I). Other studies realized in Florida, Anguilla, and New Zealand found similar lobster densities inside protected areas and in some cases even higher values than what we found in Rocas Atoll (table I). Until this moment, however, all areas studied in Brazil show less abundant populations in comparison with Rocas Atoll (Giraldes et al., 2012; Faria-Júnior et al., 2013; Gaeta et al., 2014). One probable reason may be, that these continental areas are predominantly occupied by *P. argus* and *P. laeviscauda*.

The densities found in our study were elevated, and also concentrated in a small area, so it is obviously very important to maintain this area as a marine protected area, simply because of the great abundance of lobsters and of refuges for them, as well as for other animals in this unique environment. Moreover, in this research we visualized the most abundant *P. echinatus* population in the Atlantic Ocean as compared to other places, like the archipelagos of Fernando de Noronha, the Cape Verde Islands, and the Canary Islands, which we had sampled in these last years. The same goes for other areas that have been sampled earlier by others, like Pernambuco (Giraldes et al., 2012), Abrolhos National Park (Faria-Júnior et al., 2013), the Cape Verde Islands (R. M. Freitas, pers. comm.) and the Canary Islands (Moro et al., 2014; Riera et al., 2014). Probably, this population on Rocas Atoll may serve as a source population for other areas in the Atlantic Ocean. In conclusion, protection of the unique environment of Rocas Atoll is necessary, primarily to conserve the largest and apparently best adapted population of *Panulirus echinatus* known in the world up to this moment.

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